

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 0

PHYS10342 Electricity and Magnetism

This sheet covers tutorials in weeks 1 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

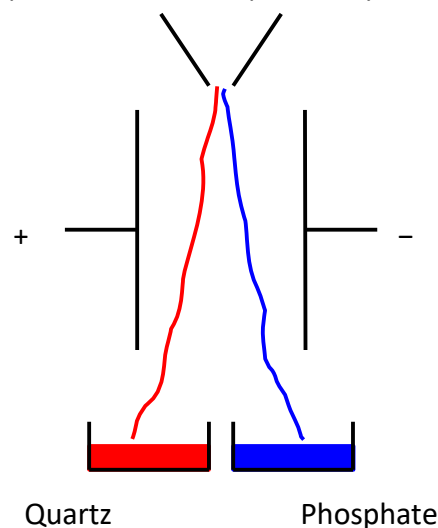
1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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E&M Tutorial Sheet 0: Charge, Coulomb's Law and Electric Forces

- 1 (Revision) Define (briefly) what is meant $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{a} \times \mathbf{b}$ for two vectors $\mathbf{a}$ and $\mathbf{b}$. Evaluate these in the case of (i) $\mathbf{a} \parallel \mathbf{b}$ ($\mathbf{a}$ and $\mathbf{b}$ parallel) (ii) $\mathbf{a} \perp \mathbf{b}$ ($\mathbf{a}$ and $\mathbf{b}$ perpendicular).
- 2 What do we mean by **charge**? Do some background reading or internet research and list the important properties of charge.
- 3 Calculate the ratio of the electrostatic repulsion to the gravitational attraction between two electrons.
Extension: assuming that charge is Lorentz invariant and mass gets corrected by a gamma factor, calculate how high a kinetic energy the electrons would need to have for the gravitational attraction to become equal to the Coulomb repulsion. Express your answer in GeV.
- 4 What magnitude uniform electric field is necessary to balance the force of gravity on paint droplets if they can be generated with a specific charge of 1 C/g? Check that you have expressed it in appropriate SI units.
When spray painting with an ordinary spray gun a huge amount of paint is wasted, up to 80% depending on the shape of the surface to be painted, draughts etc. In large scale industrial processes this would be intolerable both in economic and environmental terms. Electrostatic spraying can increase the efficiency of the painting process to nearly 100% by charging the paint droplets and applying a potential difference between the gun and the object to be coated.
- 5 Crushed phosphate ore consists of particles of quartz mixed up with particles of phosphate-bearing rock. If the mixture is vibrated, the quartz becomes negatively charged and the phosphate positively charged, each with a specific charge of $10^{-5} \text{ C kg}^{-1}$. They can therefore be electrostatically separated by letting them fall through a horizontal electric field as shown in the diagram below. If the electric field is 50 kV m^{-1} , what distance must the particles fall to be separated by a horizontal distance of 10 cm?



Hint: Analyse separately the horizontal and vertical motion. Use the horizontal motion to determine how much time you need and feed it into the vertical motion to see how far you have to fall.